

Exam Date & Time: 08-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

Computer Vision [IT-4031]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) **Examine 2D Correlation and Convolution Filters in detail. Solve the given input image (f) with its kernel (h) using convolution filtering method and display the output image (g) for the pixel values – (0, 1), (1, 2), (2, 3).**
- A)

Input Image (f)					Kernel (h)		
1	0	1	2	2	0	1	0
1	1	2	2	3	1	-4	1
1	2	2	6	3	0	1	0
1	1	2	2	3			

(5)

- B) Discuss 2D Gaussian Filtering in detail and also mention 2D Gaussian function. (3)
- C) What constitutes an image transformation, and could you elaborate on two specific transformations with examples? (2)
- 2) Identify the requirements of a good feature detector? Solve the mathematics of Harris corner detector. (5)
- A)
- B) Explain the Canny edge detector. Demonstrate the steps involved in edge detection using this detector. (3)
- C) Describe how do you detect humans using Histogram of Oriented Gradient features? (2)
- 3) Explain the organization of camera calibration. Examine the steps involved? What do we attempt to recover in camera calibration? (5)
- A)
- B) Explain & implement Epipolar Geometry in detail. What entity do you need to compute (5)

in order to use this constraint? Describe the step involved for estimating this entity.

- 4) Derive & examine the motion vectors using Lucas Kanade method. (5)
- A)
- B) Execute in detail how you would use the linear Support Vector Machine (SVM) to classify two classes that are not linearly separable. (5)
- 5) Examine the motion estimation parameters required to align successive frames in a video using KLT tracking method. (5)
- A)
- B) Explain deep learning architectures & demonstrate with respect to object recognition. (5)

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